Introduction to Small Language Models for Generative AI for Beginners

Generative AI is a fascinating field of artificial intelligence that focuses on creating systems capable of generating new content. This content can range from text and images to music and even entire virtual environments. One of the most exciting applications of generative AI is in the realm of language models.

What Are Small Language Models?

A Small Language Model (SLM) represents a scaled-down variant of a large language model (LLM), leveraging many of the architectural principles and techniques of LLMs, while exhibiting a significantly reduced computational footprint.

SLMs are a subset of language models designed to generate human-like text. Unlike their larger counterparts, such as GPT-4, SLMs are more compact and efficient, making them ideal for applications where computational resources are limited. Despite their smaller size, they can still perform a variety of tasks. Typically, SLMs are constructed by compressing or distilling LLMs, aiming to retain a substantial portion of the original model's functionality and linguistic capabilities. This reduction in model size decreases the overall complexity, making SLMs more efficient in terms of both memory usage and computational requirements. Despite these optimizations, SLMs can still perform a wide range of natural language processing (NLP) tasks:

- Text Generation: Creating coherent and contextually relevant sentences or paragraphs.
- Text Completion: Predicting and completing sentences based on a given prompt.
- Translation: Converting text from one language to another.
- Summarization: Condensing long pieces of text into shorter, more digestible summaries.

Albeit with some trade-offs in performance or depth of understanding compared to their larger counterparts.

How Do Small Language Models Work?

SLMs are trained on vast amounts of text data. During training, they learn the patterns and structures of language, enabling them to generate text that is both grammatically correct and contextually appropriate. The training process involves:

- Data Collection: Gathering large datasets of text from various sources.
- Preprocessing: Cleaning and organizing the data to make it suitable for training.
- Training: Using machine learning algorithms to teach the model how to understand and generate text.
- Fine-Tuning: Adjusting the model to improve its performance on specific tasks.

The development of SLMs aligns with the increasing need for models that can be deployed in resource-constrained environments, such as mobile devices or edge computing platforms, where full-scale LLMs may be impractical due to their heavy resource demands. By focusing on efficiency, SLMs balance performance with accessibility, enabling broader application across various domains.

Small Language Models (SLMs)

Smaller, less compute intensive models that perform well at simple tasks



Cost effective



Deployment Flexibility



Ultra-low Latency



Easier to manage and fine-tune

Learning Objectives

In this lesson, we hope to introduce the knowledge of SLM and combine it with Microsoft Phi-3 to learn different scenarios in text content, vision and MoE.

By the end of this lesson, you should be able to answer the following questions:

- What is SLM?
- What is the difference between SLM and LLM?
- What is the Microsoft Phi-3/3.5 Family?
- How to run inference with the Microsoft Phi-3/3.5 Family?

Ready? Let's get started.

The Distinctions between Large Language Models (LLMs) and Small Language Models (SLMs)

Both LLMs and SLMs are built upon foundational principles of probabilistic machine learning, following similar approaches in their architectural design, training methodologies, data generation processes, and model evaluation techniques. However, several key factors differentiate these two types of models.

Applications of Small Language Models

SLMs have a wide range of applications, including:

- Chatbots: Providing customer support and engaging with users in a conversational manner.
- Content Creation: Assisting writers by generating ideas or even drafting entire articles.
- Education: Helping students with writing assignments or learning new languages.
- Accessibility: Creating tools for individuals with disabilities, such as text-to-speech systems.

Size

A primary distinction between LLMs and SLMs lies in the scale of the models. LLMs, such as ChatGPT (GPT-4), can comprise an estimated 1.76 trillion parameters, while open-source SLMs like Mistral 7B are designed with significantly fewer parameters—approximately 7 billion. This disparity is primarily due to differences in model architecture and training processes. For instance, ChatGPT employs a self-attention mechanism within an encoder-decoder framework, whereas Mistral 7B uses sliding window attention, which enables more efficient training within a decoder-only model. This architectural variance has profound implications for the complexity and performance of these models.

Comprehension

SLMs are typically optimized for performance within specific domains, making them highly specialized but potentially limited in their ability to provide broad contextual understanding across multiple fields of knowledge. In contrast, LLMs aim to simulate human-like intelligence on a more comprehensive level. Trained on vast, diverse datasets, LLMs are designed to perform well across a variety of domains, offering greater versatility and adaptability. Consequently, LLMs are more suitable for a wider range of downstream tasks, such as natural language processing and programming.

Computing

The training and deployment of LLMs are resource-intensive processes, often requiring significant computational infrastructure, including large-scale GPU clusters. For example, training a model like ChatGPT from scratch may necessitate thousands of GPUs over extended periods. In contrast, SLMs, with their smaller parameter counts, are more accessible in terms of computational resources. Models like Mistral 7B can be trained and run on local machines equipped with moderate GPU capabilities, although training still demands several hours across multiple GPUs.

Bias

Bias is a known issue in LLMs, primarily due to the nature of the training data. These models often rely on raw, openly available data from the internet, which may underrepresent or misrepresent certain groups, introduce erroneous labeling, or reflect linguistic biases influenced by dialect, geographic variations, and grammatical rules. Additionally, the complexity of LLM architectures can inadvertently exacerbate bias, which may go unnoticed without careful fine-tuning. On the other hand, SLMs, being trained on more constrained, domain-specific datasets, are inherently less susceptible to such biases, though they are not immune to them.

Inference

The reduced size of SLMs affords them a significant advantage in terms of inference speed, allowing them to generate outputs efficiently on local hardware without the need for extensive parallel processing. In contrast, LLMs, due to their size and complexity, often require substantial parallel computational resources to achieve acceptable inference times. The presence of multiple concurrent users further slows down LLMs' response times, especially when deployed at scale.

In summary, while both LLMs and SLMs share a foundational basis in machine learning, they differ significantly in terms of model size, resource requirements, contextual understanding, susceptibility to bias, and inference speed. These distinctions reflect their respective suitability for different use cases, with LLMs being more versatile but resource-heavy, and SLMs offering more domain-specific efficiency with reduced computational demands.

Note: In this lesson, we will introduce SLM using Microsoft Phi-3 / 3.5 as an example.

Introduce Phi-3 / Phi-3.5 Family

Phi-3 / 3.5 Family mainly targets text, vision, and Agent (MoE) application scenarios:

Phi-3 / 3.5 Instruct

Mainly for text generation, chat completion, and content information extraction, etc.

Phi-3-mini

The 3.8B language model is available on Microsoft Foundry, Hugging Face, and Ollama. Phi-3 models significantly outperform language models of equal and larger sizes on key benchmarks (see benchmark numbers below, higher numbers are better). Phi-3-mini outperforms models twice its size, while Phi-3-small and Phi-3-medium outperform larger models, including GPT-3.5.

Phi-3-small & medium

With just 7B parameters, Phi-3-small beats GPT-3.5T on a variety of language, reasoning, coding, and math benchmarks.

The Phi-3-medium with 14B parameters continues this trend and outperforms the Gemini 1.0 Pro.

Phi-3.5-mini

We can think of it as an upgrade of Phi-3-mini. While the parameters remain unchanged, it improves the ability to support multiple languages (support 20+ languages: Arabic, Chinese, Czech, Danish, Dutch, English, Finnish, French, German, Hebrew, Hungarian, Italian, Japanese, Korean, Norwegian, Polish, Portuguese, Russian, Spanish, Swedish, Thai, Turkish, Ukrainian) and adds stronger support for long context.

Phi-3.5-mini with 3.8B parameters outperforms language models of the same size and is on par with models twice its size.

Phi-3 / 3.5 Vision

We can think of the Instruct model of Phi-3/3.5 as Phi's ability to understand, and Vision is what gives Phi eyes to understand the world.

Phi-3-Vision

Phi-3-vision, with only 4.2B parameters, continues this trend and outperforms larger models such as Claude-3 Haiku and Gemini 1.0 Pro V on general visual reasoning tasks, OCR, and table and diagram comprehension tasks.

Phi-3.5-Vision

Phi-3.5-Vision is also an upgrade of Phi-3-Vision, adding support for multiple images. You can think of it as an improvement in vision, not only can you see pictures, but also videos.

Phi-3.5-vision outperforms larger models such as Claude-3.5 Sonnet and Gemini 1.5 Flash across OCR, table and chart understanding tasks and on par on general visual knowledge reasoning tasks. Support multi-frame input, i.e., perform reasoning on multiple input images

Phi-3.5-MoE

Mixture of Experts(MoE) enables models to be pretrained with far less compute, which means you can dramatically scale up the model or dataset size with the same compute budget as a dense model. In particular, a MoE model should achieve the same quality as its dense counterpart much faster during pretraining.

Phi-3.5-MoE comprises 16x3.8B expert modules. Phi-3.5-MoE with only 6.6B active parameters achieves a similar level of reasoning, language understanding, and math as much larger models

We can use the Phi-3/3.5 Family model based on different scenarios. Unlike LLM, you can deploy Phi-3/3.5-mini or Phi-3/3.5-Vision on edge devices.

How to use Phi-3/3.5 Family models

We hope to use Phi-3/3.5 in different scenarios. Next, we will use Phi-3/3.5 based on different scenarios.

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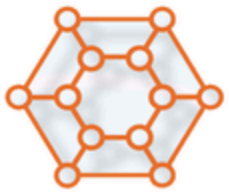
Phi-3-mini
(3.8B)



Phi-3-vision
(4.2B)



Phi-3-small
(7B)



Phi-3-medium
(14B)



Phi-3.5-mini
(3.8B)



Phi-3.5-vision
(4.2B)



Phi-3.5-MoE
(6.6B active /
42B total)

Instruction Tuned

RAI Safety Aligned

Available on





